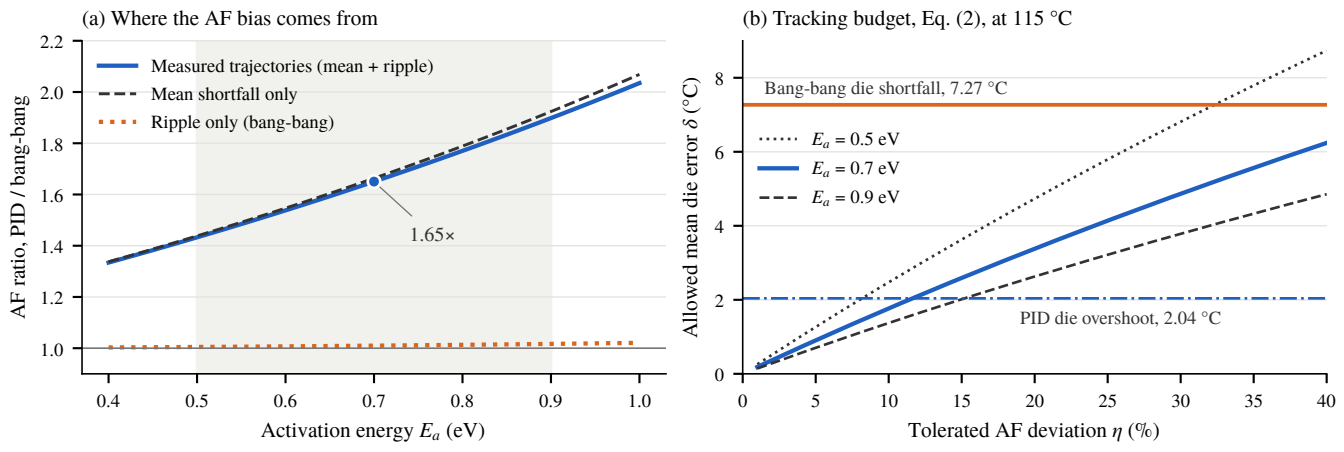
(a) Where the AF bias comes from
Measured trajectories (mean + ripple)
Mean shortfall only
Ripple only (bang-bang)
1.65×
AF ratio, PID / bang-bang
Activation energy E_a (eV)
(b) Tracking budget, Eq. (2), at 115 °C
Bang-bang die shortfall, 7.27 °C
E_a = 0.5 eV
E_a = 0.7 eV
E_a = 0.9 eV
PID die overshoot, 2.04 °C
Allowed mean die error δ (°C)
Tolerated AF deviation η (%)